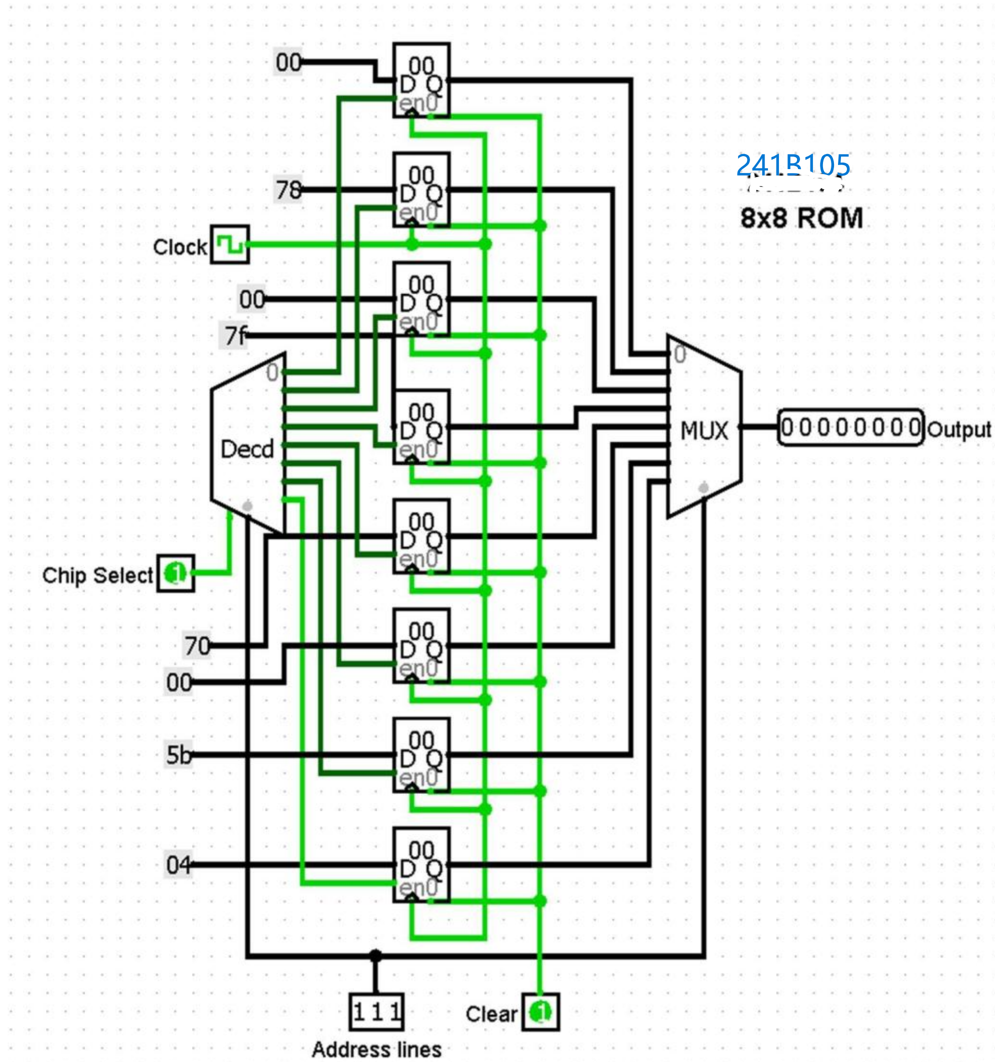
EXPERIMENT#10

Aim: Design of ROM and RAM.

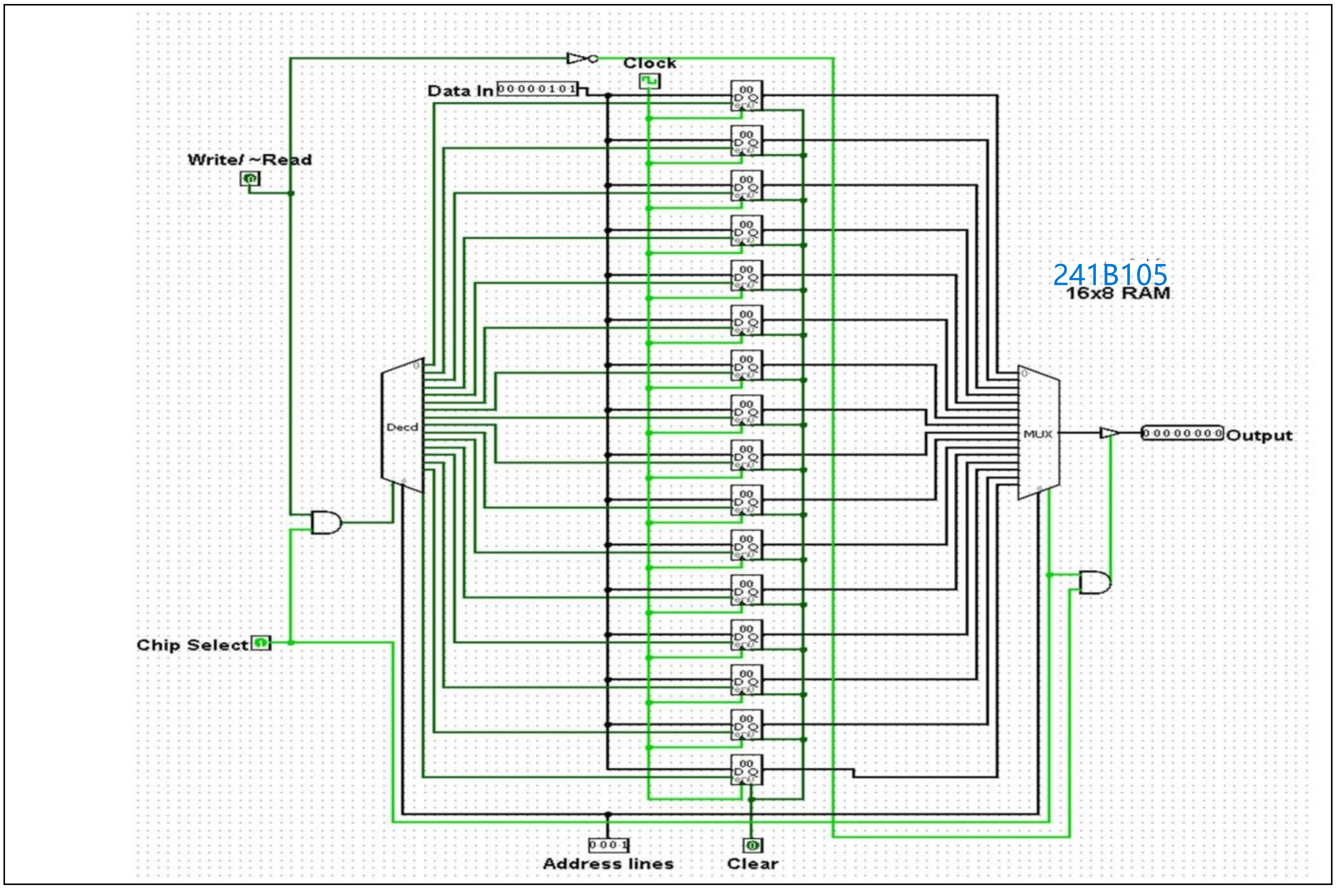
Exercise#1: Design 8×8 ROM (convert Fig. 1) and 16x8 RAM (convert to Fig. 2) using in-built block of register. Perform Write/Read operation in the memories using following in instructions:

Contents of ROM			Contents of RAM		
Address	Symbol	Hex Code	Address	Symbol	Hex Code
1	CIR	78	0	LDA	20
3	CIL	7F	3	ADD	13
4	HLT	70	A	CLA	78
6	BUN	5B	D	STA	3A
7	AND	04	E	INC	7D

8x8 ROM



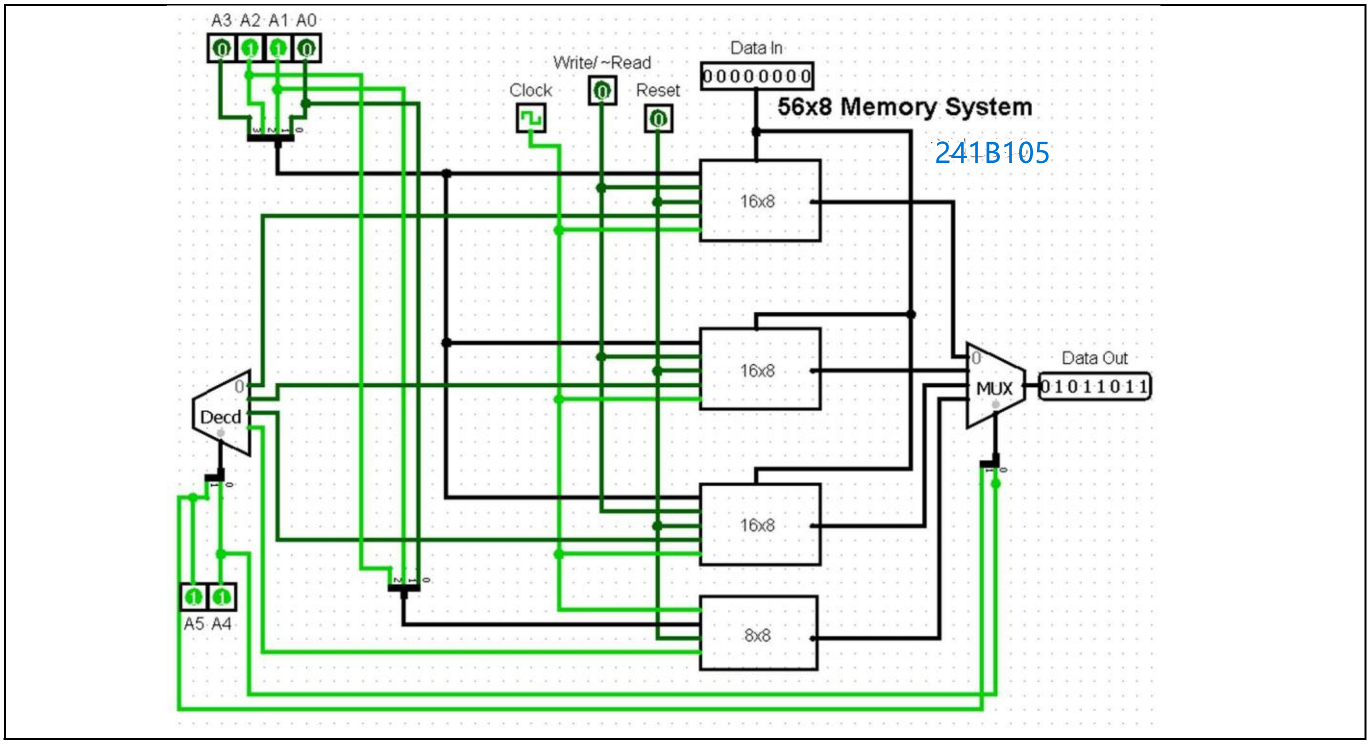
16x8 RAM



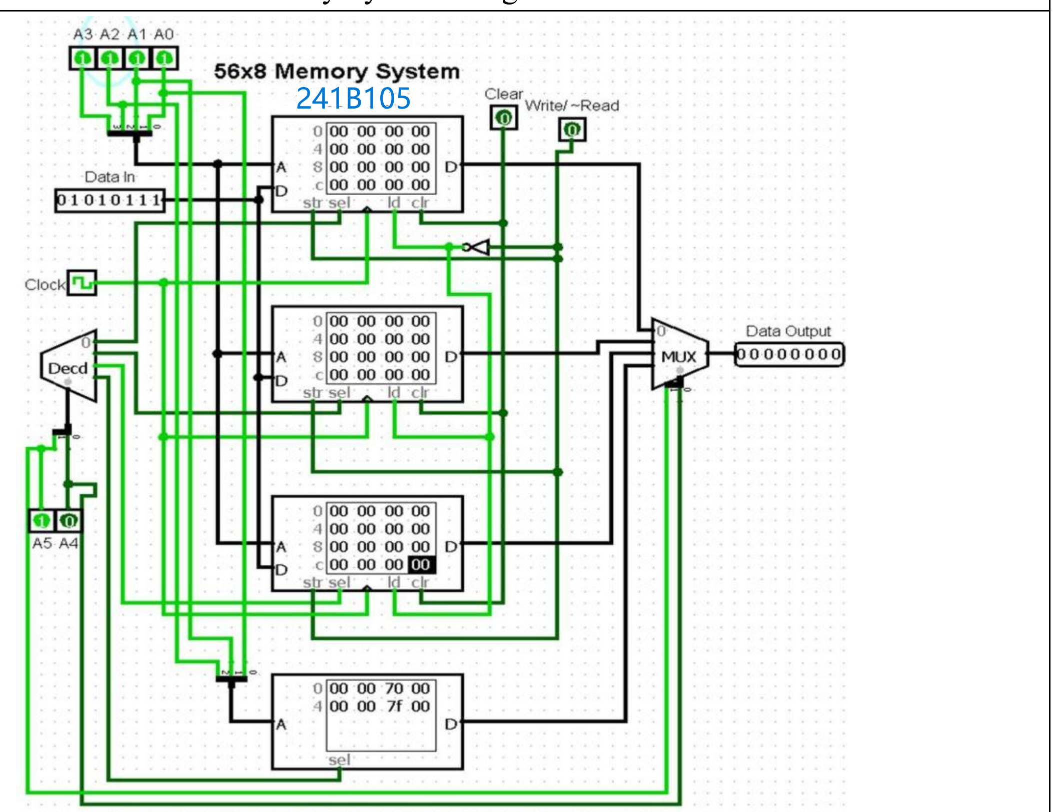
Exercise#2: Design 56×8 memory system as shown in block diagram using sub-circuits of RAM and ROM chips designed in Exercise#1. Find the address maps of each memory chip and perform Write/Read operation in the memories using instructions given in the table:

Contents of Memory			
Chip Select bits	Address bits	Symbol	Hex Code
00	0101	LDA	20
00	1101	ADD	13
01	1001	CLA	78
01	0101	STA	3A
10	1110	INC	7D
10	0000	BUN	5B
11	X111	CIL	7F
11	X100	HLT	70

X = don’t care condition

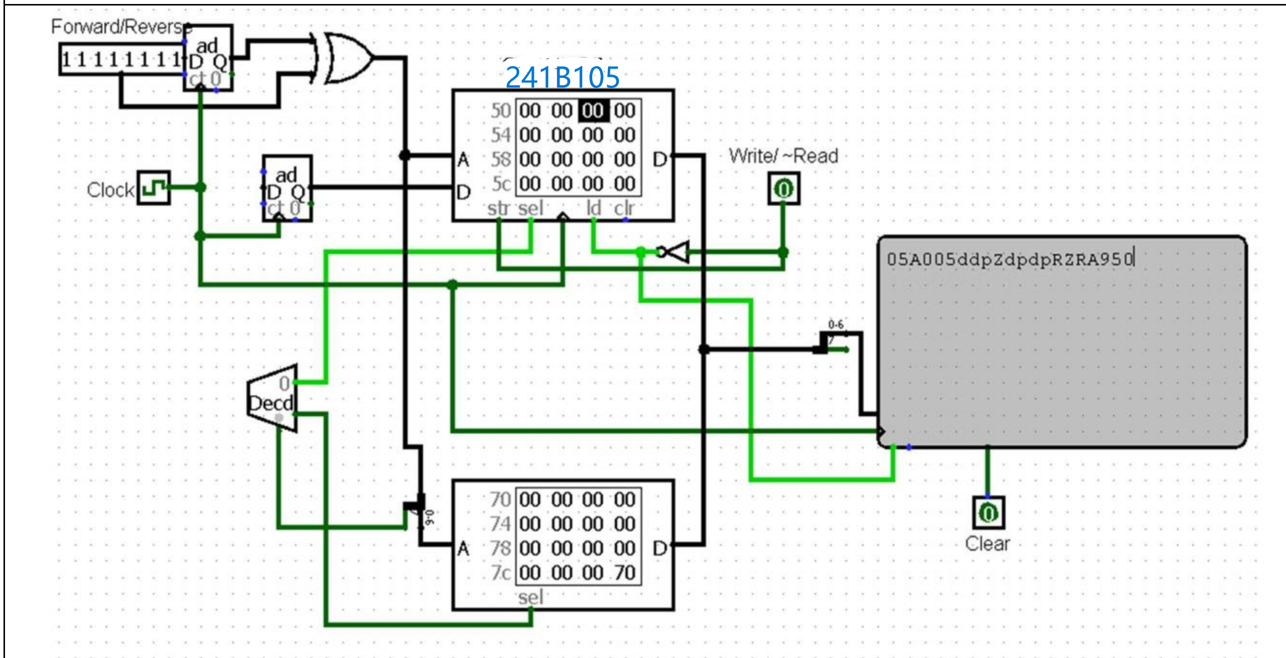


Exercise#3: Repeat Exercise#2 using in-built blocks of RAM and ROM memories. [Note: Go to RAM -> Data Interface -> Separate load and Store ports to enable data input terminal (D) and Write (str) / Read (ld) control lines.

56x8 Memory System using in-built blocks


Exercise#4: Design 384×8 memory system using in-built blocks of RAM of size 256x8 and ROM of size 128x8. Use in-built blocks of counter for auto write of data in RAM and auto address updates of both the memories. Perform following operations in the designed system:

(i) Enter Hex ASCII codes of your name in ROM.
(ii) Display the printing of your name using in-built block of TTY (teletypewriter).
(iii) Replace counter block with in-built block of Random Generator to display the printing of ASCII codes in random order.

384x8 Memory System using counters

384x8 Memory System using random generator